

Assignment - 1

	Min	Q1	Q2	Q3	Max	Q3 - Q1	1.5 x IQR)
Day	32	56	74.5	82.5	99	26.5	39.75
Night	25.5	81	81	89	98	8	12

Day

Lesser Outlier boundary (Day) = $Q1 - 1.5 \times IQR$	16.25
Greater Outlier boundary (Day) = $Q3 + 1.5 \times IQR$	122.25

- If the first data in the ordered dataset is less than 16.25 it will be less outlier. Since the minimum data 32 there are no lesser outliers.
- If the last data is more than 122.25 it will be greater outlier. But the maximum datapoint is 99. So, no greater outlier in the day-dataset.

Night

Lesser Outlier boundary (Night) = $Q1 - 1.5 \times IQR$	69
Greater Outlier boundary (Night) = $Q3 + 1.5 \times IQR$	101

- If the first data in the ordered dataset is less than 69 it will be less outlier. Since the minimum data 22.5 there are lesser outliers.
- If the last data is more than 101 it will be greater outlier. But the maximum datapoint is 98. So, no greater outlier in the night-dataset.

Assignment - 2

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5 Rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lower_Boundary	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Upper_Boundary	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1.0	40.89	37.0	50.0	50.0	51.21	200000.0
Max	215.0	89.4	97.7	91.0	98.0	77.89	940000.0
Has_Lower_Outlier	False	False	True	False	False	False	False
Has_Upper_Outlier	False	False	True	True	False	False	True

SSC_P:

Lower boundary is 37.95 and minimum value is 40.89, and upper boundary is 98.35 and its max value 89.4 is below. So, there are no outliers in SSC_P.

HSC_P:

Lower boundary is 42.75 and minimum value is 37, so this column has lower outliers, then upper boundary is 91.15 and its max value 97.7 is also beyond the upper limit. So, **there are outliers at both ends.**

Degree_P:

Lower boundary is 44.5 and minimum value is 50, and upper boundary is 88.5 and its maximum value is 91. So, **this column has the outliers in the greater end.**

eTest_P:

Lower boundary is 24.75 and its minimum value is 50 (this is fine). Then the upper boundary is 118.75 and its maximum value is 98. So, both sides are safe (no outliers)

MBA_P:

Lower boundary is 45.48 and its minimum value is 51.21 (this is fine). Then the upper boundary is 78.72 and its maximum value is 77.89. Since all values within the range no outlier replacement required.

Salary:

Lower boundary is 150k and its minimum salary is 200k, then the upper boundary is 390k and its maximum salary is 940k. Since it exceeds the upper boundary, **it needs outlier replacement.**